

AI ASSISTED CODING

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Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation

Task 1 – Portfolio Website Design

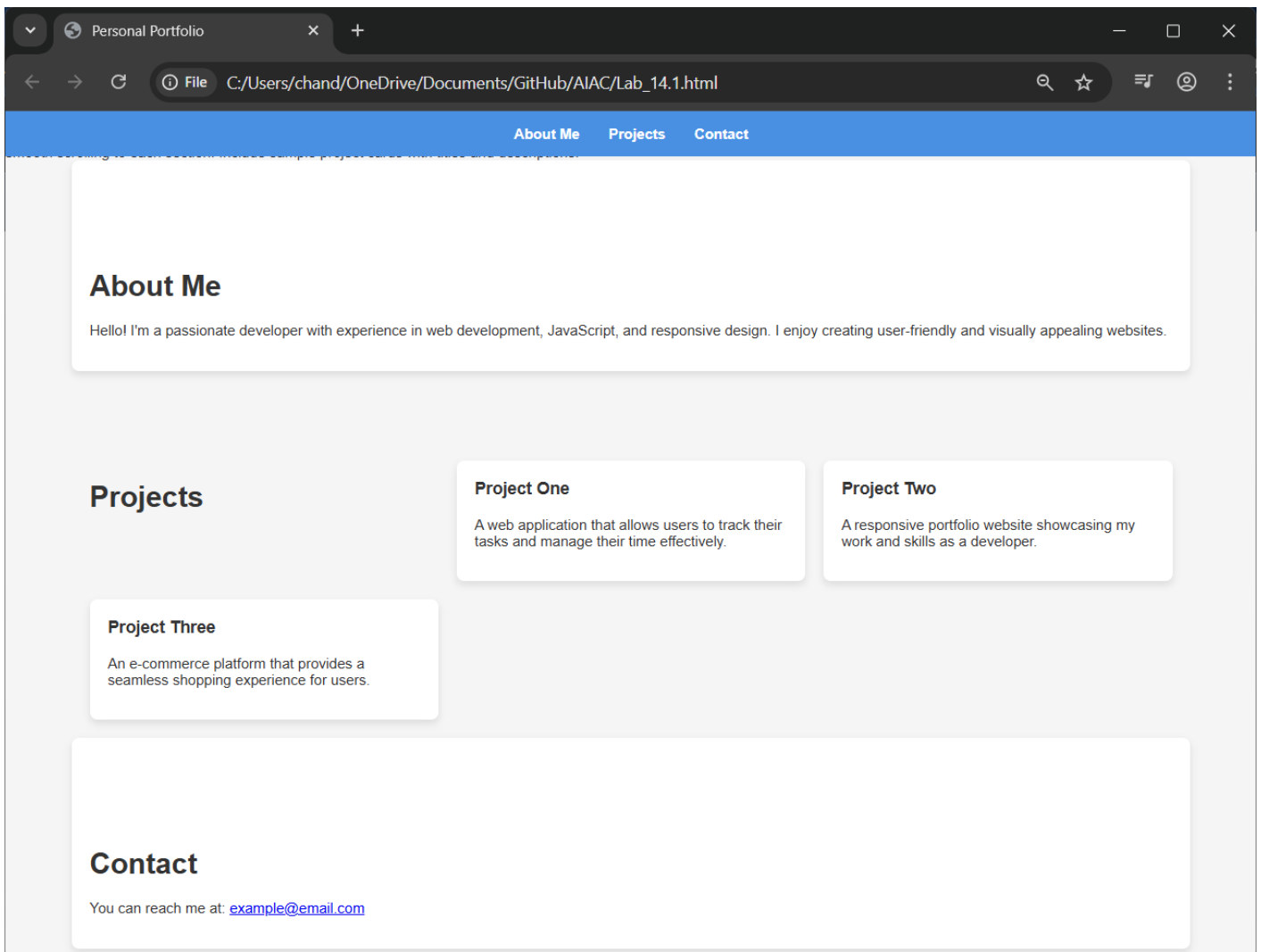
You are building a personal portfolio website to showcase your work.

Requirements:

- Create sections for About Me, Projects, and Contact.
- Use AI to:
 - o Suggest color palettes and typography.
 - o Create a responsive layout with Grid/Flexbox.
 - o Add smooth scrolling navigation.

```
Lab_14.1.html > html
1 #Task - 1
2 #Create a responsive personal portfolio website using HTML, CSS, and JavaScript. The website should include three sections: About
3 <!DOCTYPE html>
4 <html lang="en">
5 <head>
6   <meta charset="UTF-8">
7   <meta name="viewport" content="width=device-width, initial-scale=1.0">
8   <title>Personal Portfolio</title>
9   <style>
10     /* Modern color palette */
11     :root {
12       --primary-color: #4A90E2;
13       --secondary-color: #50E3C2;
14       --background-color: #F5F5F5;
15       --text-color: #333333;
16       --accent-color: #D0021B;
17     }
18
19     /* Global styles */
20     body {
21       margin: 0;
22       font-family: 'Arial', sans-serif;
23       background-color: var(--background-color);
24       color: var(--text-color);
25     }
26
27     /* Navigation bar */
28     nav {
29       position: fixed;
30       top: 0;
31       width: 100%;
32       background-color: var(--primary-color);
33       display: flex;
34       justify-content: center;
35       padding: 1rem 0;
36       z-index: 1000;
37     }
38   </style>
39 </head>
40 <body>
41   <nav>
42     <a href="#contact">Contact</a>
43   </nav>
44
45   <section id="about">
46     <h1>About Me</h1>
47     <p>Hello! I'm a passionate developer with experience in web development, JavaScript, and responsive design. I enjoy creat
48   </section>
49
50   <section id="projects">
51     <h1>Projects</h1>
52     <div class="project-card">
53       <h3>Project One</h3>
54       <p>A web application that allows users to track their tasks and manage their time effectively.</p>
55     </div>
56     <div class="project-card">
57       <h3>Project Two</h3>
58       <p>A responsive portfolio website showcasing my work and skills as a developer.</p>
59     </div>
60     <div class="project-card">
61       <h3>Project Three</h3>
62       <p>An e-commerce platform that provides a seamless shopping experience for users.</p>
63     </div>
64   </section>
65
66   <section id="contact">
67     <h1>Contact</h1>
68     <p>You can reach me at:
69     <a href="mailto:example@email.com">example@email.com</a>
70   </section>
71 </body>
72 </html>
```

```
Lab_14.1.html > html
4 <html lang="en">
95 <body>
96   <nav>
97     <a href="#contact">Contact</a>
98   </nav>
99
100   <section id="about">
101     <h1>About Me</h1>
102     <p>Hello! I'm a passionate developer with experience in web development, JavaScript, and responsive design. I enjoy creat
103   </section>
104
105   <section id="projects">
106     <h1>Projects</h1>
107     <div class="project-card">
108       <h3>Project One</h3>
109       <p>A web application that allows users to track their tasks and manage their time effectively.</p>
110     </div>
111     <div class="project-card">
112       <h3>Project Two</h3>
113       <p>A responsive portfolio website showcasing my work and skills as a developer.</p>
114     </div>
115     <div class="project-card">
116       <h3>Project Three</h3>
117       <p>An e-commerce platform that provides a seamless shopping experience for users.</p>
118     </div>
119   </section>
120
121   <section id="contact">
122     <h1>Contact</h1>
123     <p>You can reach me at:
124     <a href="mailto:example@email.com">example@email.com</a>
125   </section>
126 </body>
127 </html>
```



Task 2 – Online Store Product Page

Design a product display page for an online store.

Requirements:

- Display product image, title, price, and "Add to Cart" button.
- Use AI to:
 - o Style with BEM methodology.
 - o Make layout responsive.
 - o Add hover effects and "Add to Cart" alert.

Lab_14.1 > Lab_14.1.2.html > html > head > style > .product-card__button:hover

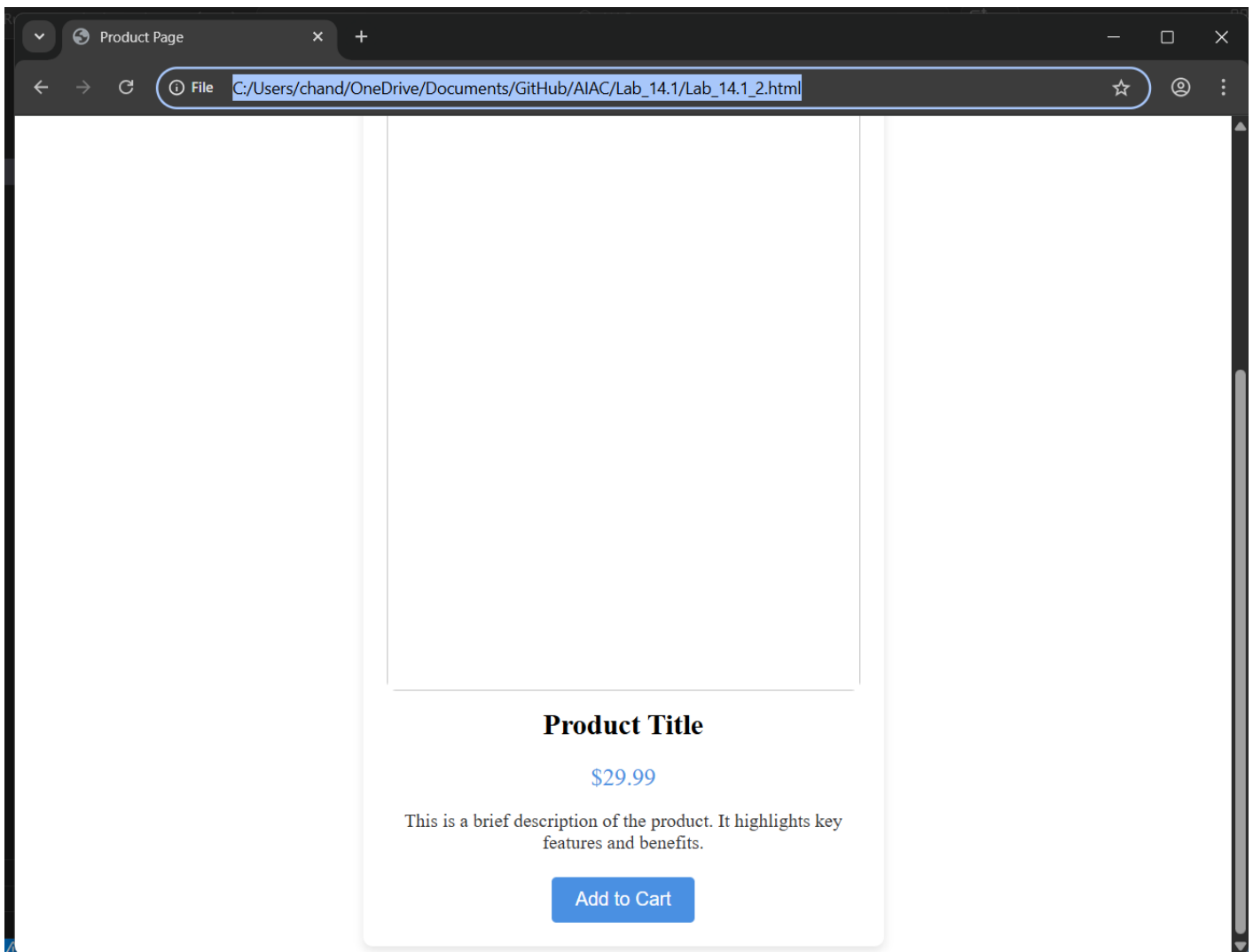


```
1 #Task - 2
2 #Create a responsive product page for an online store using HTML, CSS, and JavaScript. Display a product image, product title,
3 price, description, and an 'Add to Cart' button. Style the page using the BEM (Block Element Modifier) CSS methodology. Make
4 the layout responsive for mobile and desktop. Add hover effects to the product card and button. When the user clicks 'Add to
5 Cart', show a JavaScript alert confirming the product was added to the cart.
6 <!DOCTYPE html>
7 <html lang="en">
8 <head>
9     <meta charset="UTF-8">
10    <meta name="viewport" content="width=device-width, initial-scale=1.0">
11    <title>Product Page</title>
12    <style>
13        /* BEM CSS Methodology */
14        .product-card {
15            background-color: #fff;
16            border-radius: 8px;
17            box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
18            max-width: 400px;
19            margin: auto;
20            padding: 20px;
21            text-align: center;
22        }
23
24        .product-card__image {
25            width: 100%;
26            border-radius: 8px;
27        }
28
29        .product-card__title {
30            font-size: 1.5rem;
31            margin: 15px 0;
32        }
33
34        .product-card__price {
35            font-size: 1.25rem;
36            color: #4A90E2;
37            margin-bottom: 15px;
```

Lab_14.1 > Lab_14.1.2.html > html > head > style > .product-card__button:hover



```
7 <html lang="en">
8 <head>
12    <style>
46        .product-card__button {
54        }
55
56        .product-card__button:hover {
57            background-color: #D0021B;
58        }
59
60        /* Responsive layout */
61        @media (max-width: 600px) {
62            .product-card {
63                max-width: 100%;
64                padding: 15px;
65            }
66        }
67    </style>
68 </head>
69 <body>
70    <div class="product-card">
71        
72        <h2 class="product-card__title">Product Title</h2>
73        <p class="product-card__price">$29.99</p>
74        <p class="product-card__description">This is a brief description of the product. It highlights key features and benefits.<
75        <button class="product-card__button" onclick="addToCart()">Add to Cart</button>
76    </div>
77
78    <script>
79        function addToCart() {
80            alert('Product added to cart!');
81        }
82    </script>
83 </body>
84 </html>
```



Task 3 – Event Registration Form

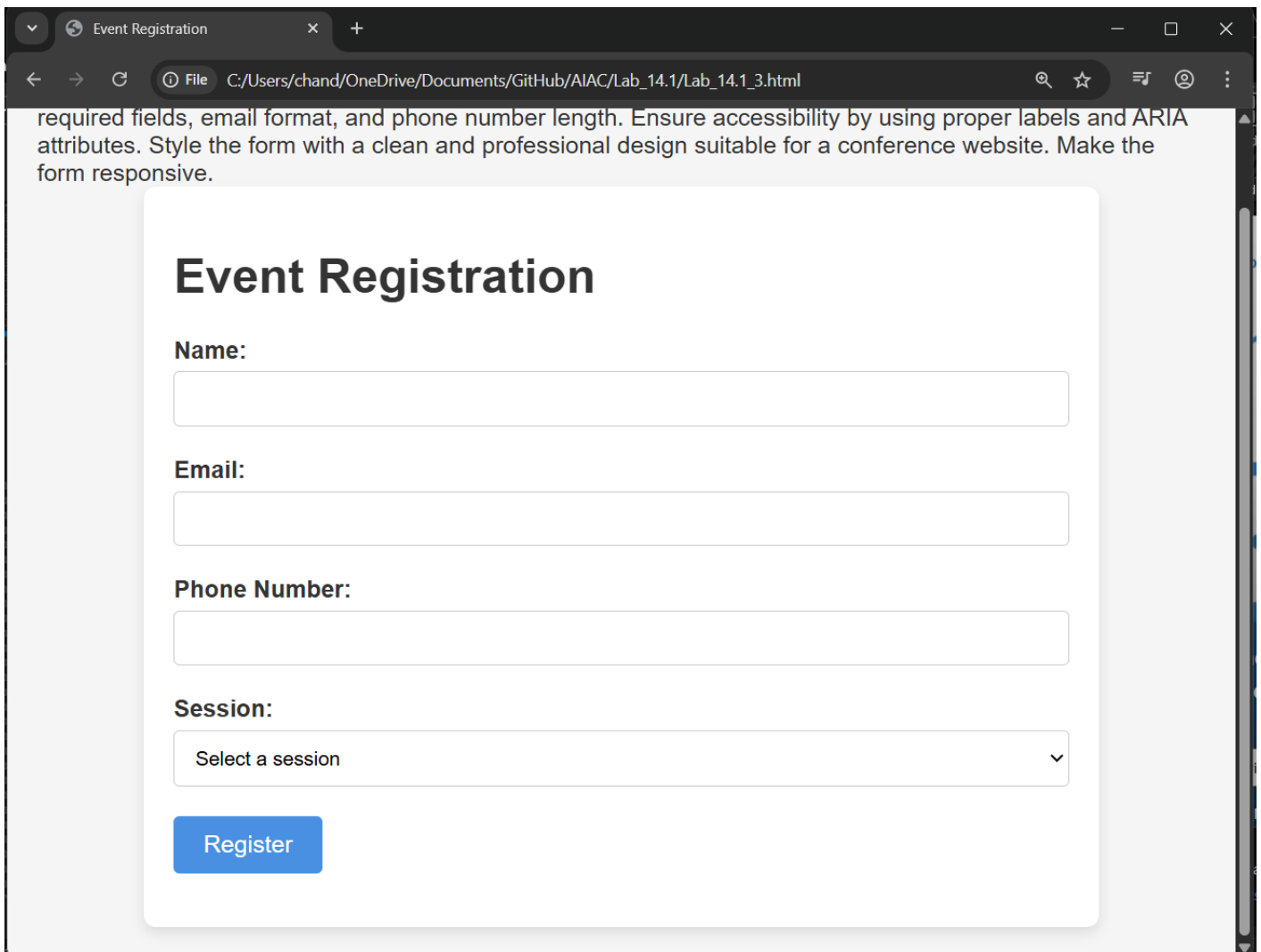
Build an event registration form for a conference.

Requirements:

- Collect name, email, phone number, and session selection.
- Use AI to:
 - o Add form validation with JavaScript.
 - o Make the form accessible with labels and ARIA.
 - o Style with a professional look.

```
Lab_14.1 > Lab_14.1_3.html > html > head > style > body
1 #Task - 3
2 #Create an event registration form using HTML, CSS, and JavaScript. The form should collect name, email, phone number, and
3 session selection using a dropdown menu. Add JavaScript form validation to check required fields, email format, and phone number
4 length. Ensure accessibility by using proper labels and ARIA attributes. Style the form with a clean and professional design
5 suitable for a conference website. Make the form responsive.
6 <!DOCTYPE html>
7 <html lang="en">
8 <head>
9   <meta charset="UTF-8">
10  <meta name="viewport" content="width=device-width, initial-scale=1.0">
11  <title>Event Registration</title>
12  <style>
13    /* Clean and professional design */
14    body {
15      font-family: 'Arial', sans-serif;
16      background-color: #f5f5f5;
17      color: #333;
18      margin: 0;
19      padding: 20px;
20    }
21
22    .form-container {
23      max-width: 600px;
24      margin: auto;
25      background-color: #fff;
26      padding: 20px;
27      border-radius: 8px;
28      box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
29    }
30
31    label {
32      display: block;
33      margin-bottom: 5px;
34      font-weight: bold;
35    }
36
37    input, select {
```

```
Lab_14.1 > Lab_14.1_3.html > html > head > style > body
7   <html lang="en">
68  <body>
94    <script>
95      document.getElementById('registrationForm').addEventListener('submit', function(event) {
96        event.preventDefault();
97
98        // Get form values
99        const name = document.getElementById('name').value.trim();
100       const email = document.getElementById('email').value.trim();
101       const phone = document.getElementById('phone').value.trim();
102       const session = document.getElementById('session').value;
103
104       // Simple validation
105       if (!name || !email || !phone || !session) {
106         alert('Please fill in all required fields.');
```



Task 4 – Resume Website

Build a one-page online resume.

Requirements:

- Sections: Profile Summary, Skills, Education, Experience.
- Use AI to:
 - o Propose professional typography styles.
 - o Align sections using CSS Grid.
 - o Add smooth scroll navigation between sections.

Lab_14.1 > 5 Lab_14.1_4.html > ...



```

1 #Task - 4
2 #Create a one-page resume with Profile, Skills, Education, and Experience. Use CSS Grid, professional fonts,
3 and smooth scrolling navigation.
4 <!DOCTYPE html>
5 <html lang="en">
6 <head>
7   <meta charset="UTF-8">
8   <meta name="viewport" content="width=device-width, initial-scale=1.0">
9   <title>One-Page Resume</title>
10  <style>
11    /* Professional fonts */
12    @import url('https://fonts.googleapis.com/css2?family=Roboto:wght@400;700&display=swap');
13
14    body {
15      font-family: 'Roboto', sans-serif;
16      background-color: #f5f5f5;
17      color: #333;
18      margin: 0;
19      padding: 20px;
20    }
21
22    .resume-container {
23      max-width: 800px;
24      margin: auto;
25      background-color: #fff;
26      padding: 20px;
27      border-radius: 8px;
28      box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
29    }
30
31    h1, h2 {
32      color: #4A90E2;
33    }
34
35    /* CSS Grid layout */
36    .grid-container {
37      display: grid;

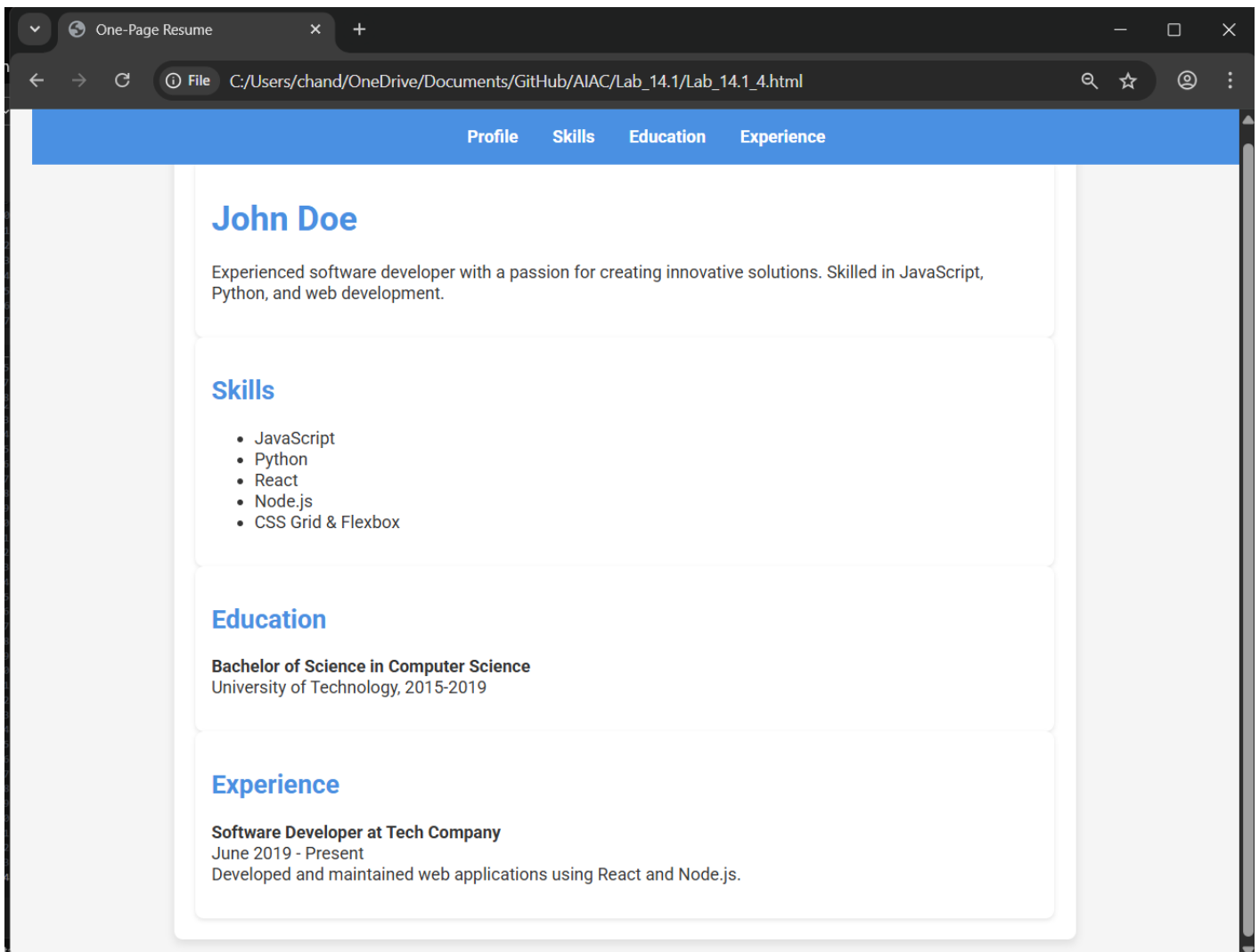
```

Lab_14.1 > Lab_14.1_4.html > ...

```

5 <html lang="en">
77 <body>
78   <nav>
79     <a href="#experience">Experience</a>
83   </nav>
84
85   <div class="resume-container">
86     <section id="profile" class="section">
87       <h1>John Doe</h1>
88       <p>Experienced software developer with a passion for creating innovative solutions. Skilled in JavaScript, Python, and
89     </section>
90
91     <section id="skills" class="section">
92       <h2>Skills</h2>
93       <ul>
94         <li>JavaScript</li>
95         <li>Python</li>
96         <li>React</li>
97         <li>Node.js</li>
98         <li>CSS Grid & Flexbox</li>
99       </ul>
100     </section>
101
102     <section id="education" class="section">
103       <h2>Education</h2>
104       <p><strong>Bachelor of Science in Computer Science</strong><br>University of Technology, 2015-2019</p>
105     </section>
106
107     <section id="experience" class="section">
108       <h2>Experience</h2>
109       <p><strong>Software Developer at Tech Company</strong><br>June 2019 - Present<br>Developed and maintained web applications
110     </section>
111   </div>
112 </body>
113 </html>
114

```

Task 5 – Restaurant Menu Page

Design a digital menu page for a restaurant.

Requirements:

- Display menu items with name, description, and price.
- Use AI to:
 - o Suggest a food-friendly color palette.
 - o Design a responsive card-based layout.
 - o Add hover effects on menu items.

```
Lab_14.1 > Lab_14.1_5.html > ...
1 #Task - 5
2 #Create a responsive restaurant menu page with item name, description, and price. Use card layout,
3 | food-style colors, and hover effects.
4 <!DOCTYPE html>
5 <html lang="en">
6 <head>
7   <meta charset="UTF-8">
8   <meta name="viewport" content="width=device-width, initial-scale=1.0">
9   <title>Restaurant Menu</title>
10  <style>
11    /* Food-style colors */
12    body {
13      font-family: 'Arial', sans-serif;
14      background-color: #fff8e1;
15      color: #333;
16      margin: 0;
17      padding: 20px;
18    }
19
20    .menu-container {
21      max-width: 1200px;
22      margin: auto;
23      display: grid;
24      grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
25      gap: 20px;
26    }
27
28    .menu-item {
29      background-color: #fff3e0;
30      border-radius: 8px;
31      box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
32      padding: 20px;
33      text-align: center;
34      transition: transform 0.3s ease, box-shadow 0.3s ease;
35    }
36
37    .menu-item:hover {
```

